

PROGRAM B: BEST FIT

C PROGRAM :

```
#include <stdio.h>
```

```
int main() {
```

```
    int blockSize[20], processSize[20];
```

```
    int allocation[20];
```

```
    int nb, np, i, j, bestIdx;
```

```
    printf("Enter Number of Blocks: ");
```

```
    scanf("%d", &nb);
```

```
    printf("Enter Number of Processes: ");
```

```
    scanf("%d", &np);
```

```
    printf("Enter Block Sizes:\n");
```

```
    for (i = 0; i < nb; i++) {
```

```
        scanf("%d", &blockSize[i]);
```

```
    }
```

```
    printf("Enter Process Sizes:\n");
```

```
    for (i = 0; i < np; i++) {
```

```
        scanf("%d", &processSize[i]);
```

```
    }
```

```
    for (i = 0; i < np; i++) {
```

```
        allocation[i] = -1;
```

```
    }
```

```
    for (i = 0; i < np; i++)
```

```
    {
```

```
        bestIdx = -1;
```

```
        for (j = 0; j < nb; j++) {
```

```
            if (blockSize[j] >= processSize[i]) {
```

```
                if (bestIdx == -1 || blockSize[j] < blockSize[bestIdx]) {
```

```
                    bestIdx = j;
```

```

        }
    }
}
if (bestIdx != -1) {
    allocation[i] = bestIdx;
    blockSize[bestIdx] -= processSize[i];
}
}
printf("\nProcess No\tProcess Size\tBlock No\n");
for (i = 0; i < np; i++) {
    printf("%d\t\t%d\t\t", i + 1, processSize[i]);
    if (allocation[i] != -1) {
        printf("%d\n", allocation[i] + 1);
    }
    else {
        printf("Not Allocated\n");
    }
}
return 0;
}

```

SHELL SCRIPT :

```

GNU nano 8.7.1
#!/bin/bash

echo "Best Fit Memory Allocation Demonstration"

blocks=(100 500 200 300 600)
processes=(212 417 112 426)

echo "Memory Blocks: ${blocks[@]}"
echo "Processes: ${processes[@]}"

echo "Allocation Performed Using Best Fit"

```

Output :

```
(ganesh@LAPTOP-R3RDH9NG)~  
$ nano bestfit.sh  
  
(ganesh@LAPTOP-R3RDH9NG)~  
$ chmod +x bestfit.sh  
  
(ganesh@LAPTOP-R3RDH9NG)~  
$ bash bestfit.sh  
Best Fit Memory Allocation Demonstration  
Memory Blocks: 100 500 200 300 600  
Processes: 212 417 112 426  
Allocation Performed Using Best Fit
```